Question 1 (30 pts)

In order to separate signals of different frequencies, Fourier transform and bandpass filter are applied. Numpy is used as np.fft.rfft for Fourier transform. From frequency-domain plots, the original signal consists of 2 signals of 150 Hz and 700 Hz. 150Hz falls into the human-voice interval, and 700Hz falls into the instrument's interval. By transforming time-domain to frequency-domain, and then filtering, they can be easily separated. These separated signals again transformed into time-domain(IFFT) and saved in .wav files.

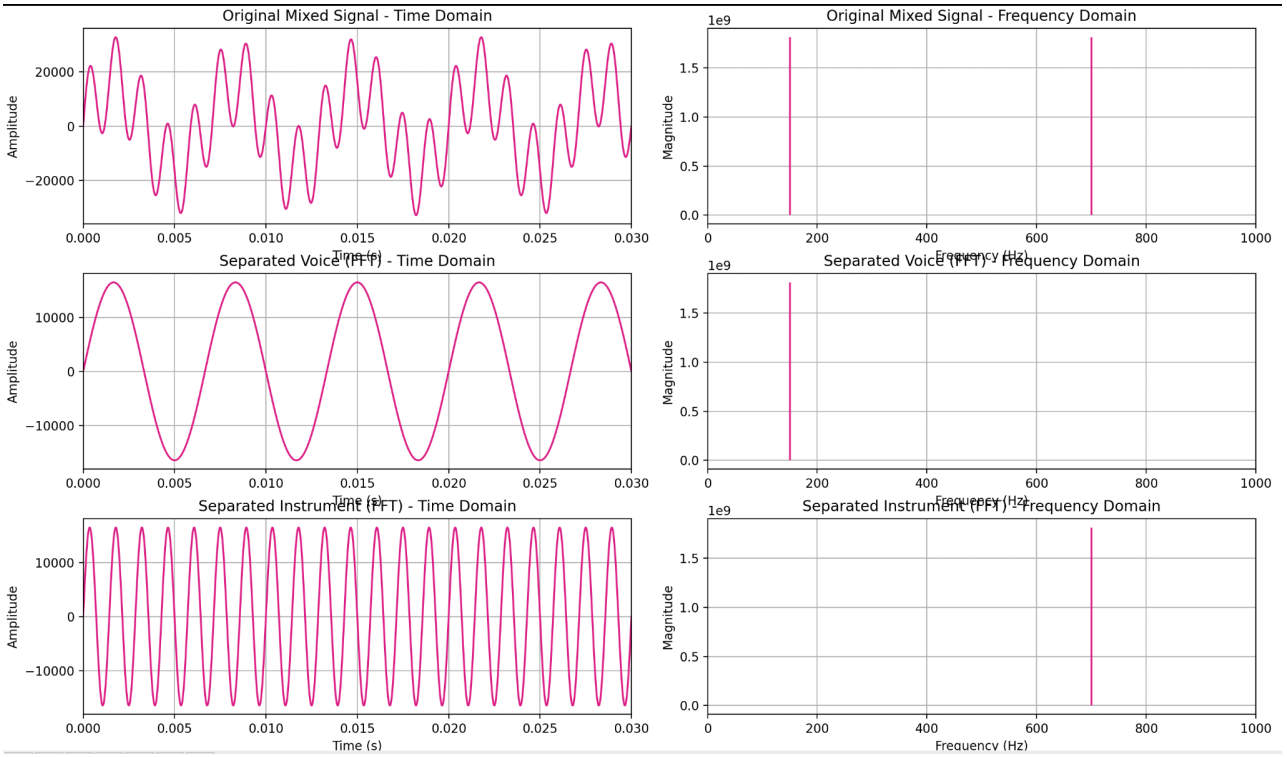

Figure 1. Plots of Question 1

Question 2 (30 pts)

Combining two signals of different frequencies is done by summing separate parts. From Figure 2, it is seen that the combined signal is the same as the original one:

- Frequencies are same as original: 150Hz and 700 Hz
- In time-domain, overall waveforms are the same too.

This is because there are only 2 frequencies which are within filtering bandwidth, and no frequency is masked out with zeros during filtering. So when combining back, the original signal is obtained.

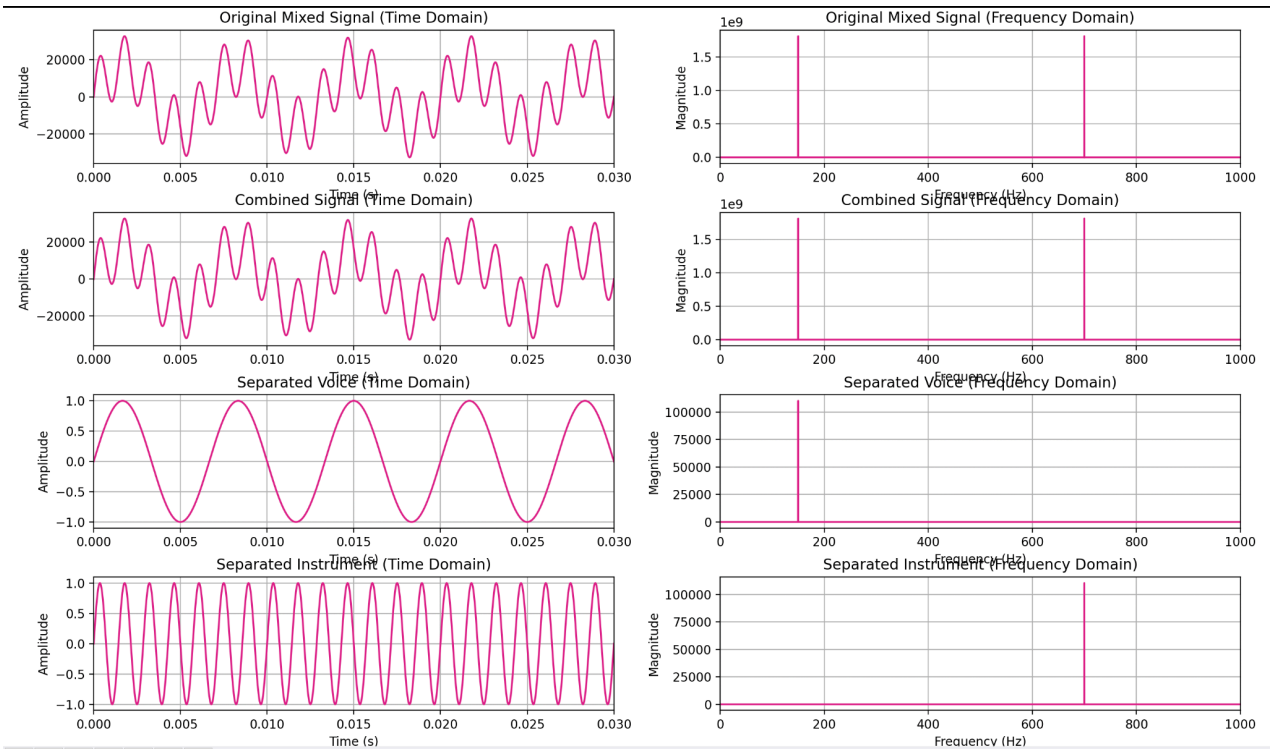

Figure 2. Plots of Question 2

Question 3 (40 pts)

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\left(\frac{2\pi}{N}\right)kn}, \qquad k = 0, 1, \ldots, N-1$$

Figure 3. DFT formula

The DFT formula is used in the DFT function in my code, it transforms time-domain to frequency-domain samples. Feature of a signal is an array of magnitudes of each sample at each frequency. First, sine, square and triangle waves' features are extracted using 4-point and 8-point DFT. All the unknown samples follow the same procedure. Then I compared unknown signals' features to known ones by calculating the norm of an array(difference between every point). The one that produces the minimum norm is classified as one of sine, square or triangle waves. Figure 4 shows classification of 12 samples given in the task.

```
(.venv) madinaalzhanova@Madinas-Mac BLG354E_HW2 % python3 question3.py
=== Classifications for 12 Samples ===
Sample 1 is sine wave, Distances = {'sine': np.float64(0.5241966836824695), 'square': np.float64(6.763540904931084), 'triangle': np.float64(7.132778173173256)}
Sample 2 is sine wave, Distances = {'sine': np.float64(1.03577658128188), 'square': np.float64(6.296856555571612), 'triangle': np.float64(6.751452611637296)}
Sample 3 is sine wave, Distances = {'sine': np.float64(1.5310887292943038), 'square': np.float64(5.850201146894507), 'triangle': np.float64(6.396298454380035)}
Sample 4 is sine wave, Distances = {'sine': np.float64(2.006321711389396), 'square': np.float64(5.4270793707790315), 'triangle': np.float64(6.069825589875475)}
Sample 5 is square wave, Distances = {'sine': np.float64(7.246873286926362), 'square': np.float64(0.0), 'triangle': np.float64(3.002447356483745)}
Sample 6 is square wave, Distances = {'sine': np.float64(7.246873286926362), 'square': np.float64(0.0), 'triangle': np.float64(3.002447356483745)}
Sample 7 is square wave, Distances = {'sine': np.float64(7.246873286926362), 'square': np.float64(0.0), 'triangle': np.float64(3.002447356483745)}
Sample 8 is square wave, Distances = {'sine': np.float64(7.246873286926362), 'square': np.float64(0.0), 'triangle': np.float64(3.002447356483745)}
Sample 9 is triangle wave, Distances = {'sine': np.float64(6.992849823135724), 'square': np.float64(2.6392666060580146), 'triangle': np.float64(0.6842619991564801)}
Sample 10 is triangle wave, Distances = {'sine': np.float64(6.730121727335588), 'square': np.float64(2.508541421025921), 'triangle': np.float64(1.0264428755749104)}
Sample 11 is triangle wave, Distances = {'sine': np.float64(6.474778876377886), 'square': np.float64(2.4195031864515295), 'triangle': np.float64(1.368573925816957)}
Sample 12 is triangle wave, Distances = {'sine': np.float64(7.262259709787797), 'square': np.float64(2.8059071189020925), 'triangle': np.float64(0.3421239712336811)}
```

Figure 4. Signal classification results

Figure 5 is a plot of all samples in the frequency domain. Sine waves look like a fragment of a sine wave, square waves have one main frequency, others are of equal less magnitude. Triangle waves' feature resemble square waves' feature of having one peak frequency, however other points'

magnitude is very low. According to these features the samples given are classified into categories.

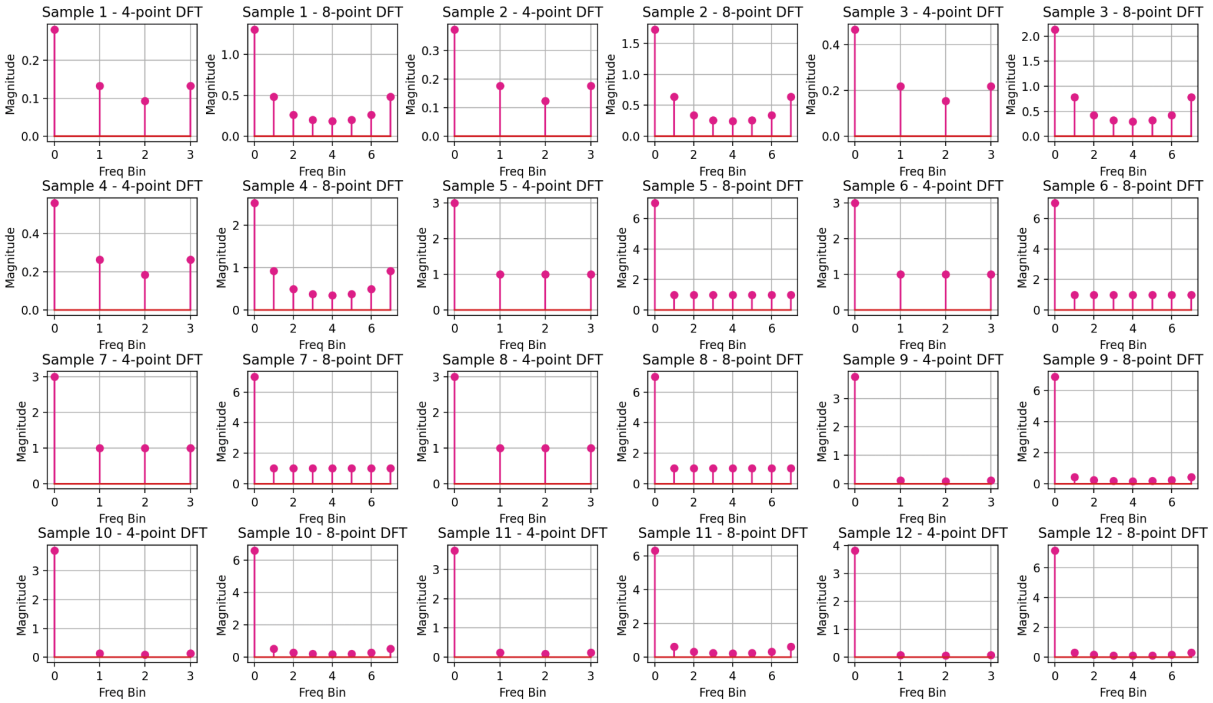

Figure 5. Plot for Question 3